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Cargo Unit

Abstract

Described herein are units for transportation of cargo, and methods of using them. In some embodiments, the units are collapsible, stackable cargo units. In some embodiments, the cargo units include a generally rectangular base having a front, a rear and two sides, with forklift slots on its front, rear and each side. The cargo units may further include front and rear walls extending upward from the base. The cargo units may further include first and second side walls extending upward from the base between the front and rear walls. The first side wall may have a length greater than the second side wall. The cargo units may further include an integrated lid hingedly attached to one or more of the front, rear, and side walls.

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Background/Summary

CROSS-REFERENCE TO RELATED APPLICATIONS [0001] This application is a continuation of prior application Ser. No. 18/529,849, filed Dec. 5, 2023, which is a continuation of prior application Ser. No. 17/434,366, filed Aug. 26, 2021, now U.S. Pat. No. 11,866,250 B2, which is a U.S. National Phase Application filed under 35 U.S.C. § 371 of International Application Number PCT/IB2020/000183, filed Mar. 4, 2020, designating the United States, which claims benefit of U.S. Provisional Application No. 62/813,369, filed Mar. 4, 2019, which are hereby fully incorporated by reference as if fully set forth herein.

FIELD

[0002] This relates generally to units for transportation of cargo, and more particularly to collapsible, stackable cargo units.

BACKGROUND

[0003] Reusable, stackable containers such as those disclosed in U.S. Pat. No. 8,573,427, the disclosure of which is incorporated herein by reference, can be useful for transporting various types of cargo, e.g., by rail, in overseas commerce, and in other contexts. Where a fleet of containers of this type is intended to include stacks of up to, e.g., four or five fully-loaded containers during shipping operations, it is desirable for each container in a fleet to be capable of supporting the weight of multiple fully loaded containers stacked above it. Other prior stackable containers are described in CN103158924B and CN201172503Y.

[0004] A problem with some containers is that components such as lids, removable walls and the like can be separated from other components during use. Permanently connecting all of components together might potentially solve this issue, but removal of lids and walls can sometimes be useful in facilitating loading and unloading.

[0005] Another problem associated with collapsible shipping containers is that they can be damaged during use, in erected or collapsed configurations, e.g., by impacts with other containers and forklift tines. A further problem is that the cargo may exert very high outward pressure on the sidewalls. One particular material that can apply significant outward pressure on the sidewalls is rubber, which in some cases can apply increasing outward pressure over time during transport. Significant inward pressure and impact loads may also be applied to the sidewalls by forklift tines and by other containers during handling. For example, when containers are shipped by rail, stopping and starting of railcars may result in adjacent containers contacting, impacting and/or otherwise exerting forces on one another.

[0006] To provide the strength required to withstand such pressures as well as stacking loads and other loads associated with, e.g., handling by forklifts, many commercial shipping containers have a relatively heavy steel construction, and as a result their weight may be a significant percentage of their capacity. For example, a Goodpack MB5 intermediate bulk container that is capable of carrying a payload of about 1,650 kg. or 3,637 lbs. in commercial use may have a tare weight of about 126 to 136 kg., or 280 to 300 lbs. There is a continuing interest in providing new and improved shipping containers that provide similar functionality to existing commercial containers but with decreased tare weight.

[0007] Collapsible shipping containers often include a bar code label, RFID tag, or other means to facilitate tracking by providing a unique, machine readable identifier for each container. In some cases, labels or tags can be difficult to access, e.g., for containers at the top of a five-high stack.

Such labels and tags can also be subject to damage or wear due, e.g., to impacts with or abrasion by forklift tines, other containers. There is a need for improvement in this area as well.

SUMMARY

[0008] Disclosed herein is a collapsible, stackable cargo unit having an erected configuration in which the cargo unit may define an enclosed interior, and a collapsed configuration in which the cargo unit may be stored and transported compactly. The cargo unit comprises a generally rectangular base having a front, a rear and two sides, with forklift slots on its front, rear and each side; front and rear walls extending upward from the base; first and second side walls extending upward from the base between the front and rear walls; and an integrated lid hingedly attached to one or more of the front, rear and side walls.

[0009] In some embodiments, all four walls are fully removable.

[0010] In some embodiments, the integrated lid comprises a first lid member that includes a hinged edge hingedly attached to the first side wall, and a second lid member that includes a hinged edge hingedly attached to the second side wall. In some embodiments, each of the first and second lid members includes a free edge opposite its hinged edge, wherein the first lid member is larger than the second lid member, with a dimension between its hinged edge and its free edge being larger than a corresponding dimension of the second lid member. In some embodiments, each lid member has a first handle extending outward over the front wall, and a second handle extending outward over the rear wall, to facilitate lifting of the lid to open the cargo unit. Each handle may include a downwardly-projecting end that wraps downward over the top of its associated front or rear wall to add strength and rigidity to the structure, and/or to help maintain the lid members in alignment when in closed position.

[0011] In some embodiments, each of the lid members is permanently attached to a side wall, and is capable of pivoting through a 270° arc between a generally horizontal closed position in which the free ends of the lid members are adjacent one another, and a generally vertical open position in which the lid member hangs against an outside surface of its associated side wall, when the cargo unit is erected.

[0012] In some embodiments, the front wall comprises releasable locking mechanisms that are capable of locking the cargo unit in erected configuration. In some embodiments, the front wall comprises a lower front wall and an upper front wall pivotably connected thereto, wherein the lower front wall includes locking mechanisms, the upper front wall includes locking mechanisms, and when the cargo unit is in its erected configuration, the upper front wall may be either (1) locked in an upright configuration associated with a fully closed condition of the cargo unit, or (2) unlocked and folded outward and down through an arc of about 180° to hang against an outer surface of the lower front wall while the lower front wall remains locked in place to partially open the cargo unit and facilitate access to the interior. In some embodiments, the rear wall is identical or substantially identical to the front wall as described above.

[0013] In some embodiments, each of the base and walls comprises a metal frame, and one or more generally flat, planar metal sheets or plates welded or otherwise attached to the frame. The metal may be, e.g., galvanized steel, and the cargo unit may have a tare weight of, e.g., 70 to 100 kg., 80 to 90 kg., or about 84 kg. In other embodiments, one or more of the base and walls comprises a unitary, one-piece stamped primary member incorporating integral structural contours to provide strength and rigidity, instead of comprising a frame combined with sheets or plates, and the cargo unit may have a tare weight significantly less than 80 kg.

[0014] Also disclosed herein is a double cargo unit comprising a pair of similar individual cargo units joined to one another back-to-back, wherein each of the individual cargo units is similar to one of those described above, except that it does not include a rear wall. Thus, the individual cargo units in these embodiments are joined at their respective open sides. The individual cargo units may be joined by bridge plates, splice bars, or other structures joining one or more of their base members, side walls and lids together, and/or by other means such as bolting adjacent portions of

the bases directly to one another. In some embodiments, the bases together define a generally flat floor, without any significant upward projection or rim protruding upward at the joint between the bases of the individual cargo units. In some embodiments, increased stiffness and strength are provided by a long, narrow structural member such as a steel plate, steel angle, or a tubular member of e.g., rectangular cross section, extends across the joint at lid height, joining all four of the upper side wall corners adjacent the joint, and providing a support for edges of the lid members adjacent the joint. This structural member may have a slot or recess to accommodate downward projections of the lid handles.

[0015] In some embodiments, the integrated lid of each individual cargo unit comprises a first lid member that includes a hinged edge hingedly attached to the first side wall, and a second lid member that includes a hinged edge hingedly attached to a second side wall.

[0016] Also disclosed herein is a method of unloading and collapsing a collapsible, stackable cargo unit as described above having an erected configuration in which the cargo unit defines an enclosed interior, and a collapsed configuration in which the cargo unit may be stored and transported compactly, the cargo unit comprising a generally rectangular base having a front, a rear and two sides, with forklift slots on its front, rear and each side; front and rear walls extending upward from the base; first and second side walls extending upward from the base between the front and rear walls; locking mechanisms that are capable of holding the cargo unit in its erected configurations; and a two-piece integrated lid that is in a generally horizontal position when the cargo unit is in its erected configuration.

[0017] In some embodiments, the method may comprise: [0018] a. opening first and second lid members by pivoting each lid member through an arc of about 270° about a hinge that connects it to a respective one of the first and second side walls; [0019] b. removing cargo from the cargo unit; [0020] c. releasing locking mechanisms on the front wall, and pivoting the front wall downward and inward about a pivotable connection to the base through an arc of about 90° so that it rests on the base; [0021] d. releasing locking mechanisms on the rear wall, and pivoting the rear wall downward and inward about a pivotable connection to the base through an arc of about 90° so that it rests on the base, with the pivotable connection comprising an upwardly extending slot on the base that permits upward movement of the pivotable connection of the rear wall to the base, with the pivot axis of the rear wall being fixed at an elevation that is vertically offset above that of the front wall by a dimension equal or about equal to the thickness of the front wall, and with the side walls remaining upright, supported by base corner posts; [0022] e. pivoting a first side wall downward and inward about a pivotable connection to the base through an arc of about 90° so that it rests on the rear wall, with the pivot axis of the first side wall being fixed at an elevation that permits the first side wall to lay flat in an approximately horizontal orientation on the rear wall; [0023] f. pivoting a second side wall downward and inward about a pivotable connection to the base through an arc of about 90° so that it rests on the first side wall, with the pivot axis of the second side wall being fixed at an elevation that permits the second side wall to lay flat in an approximately horizontal orientation on the first side wall; and [0024] g. pivoting the second lid member through an arc of about 180° about the hinge that connects it to the second side wall so that the second lid member rests on the first lid member rather than the second side wall, thereby reducing the height of the collapsed cargo unit.

[0025] In some embodiments, methods of erecting a collapsed cargo unit may comprise reversing the above steps. In some embodiments, methods of collapsing or erecting a double cargo unit as described above may include methods identical to those described above, except that the steps involving rear walls will not be applicable, and the methods involving the other walls will need to be carried out on each of the individual cargo units.

[0026] In some embodiments involving loading or unloading of the individual or double cargo units described above, any number of walls may be removed from the base prior to loading or unloading. This may facilitate access to the interior from one or more sides, which may be useful in

some contexts involving, e.g., manual loading or unloading, and/or use of forklifts to load or unload.

[0027] It is contemplated that in various embodiments, any number of the features described herein may be included in combination with one another, to the extent practical. It is further contemplated that in various embodiments, any number of the features described in the above-mentioned U.S. Pat. No. 8,573,427 may be included in combination with one or more of the features described herein, to the extent practical.

Description

BRIEF DESCRIPTION OF FIGURES

[0028] FIG. 1 is a perspective view of an individual cargo unit in an erect configuration.

[0029] FIG. 2 is a perspective view of the individual cargo unit of FIG. 1 in a collapsed configuration.

[0030] FIG. 3 is a perspective view of a double cargo unit in an erect configuration.

[0031] FIG. 4 is a perspective view of the double cargo unit of FIG. 3 in a collapsed configuration.

[0032] FIG. 5A is a perspective view of the individual cargo unit of FIG. 1 with the lid members rotated to open positions.

[0033] FIG. 5B is a perspective view of the individual cargo unit of FIG. 1 with the front wall in a collapsed position.

[0034] FIG. 5C is a perspective view of the individual cargo unit of FIG. 1 with the rear wall in a collapsed position.

[0035] FIG. 5D is a perspective view of the individual cargo unit of FIG. 1 with the first side wall in a collapsed position.

[0036] FIG. 5E is a perspective view of the individual cargo unit of FIG. 1 with the second side wall in a collapsed position.

[0037] FIG. 5F is a perspective view of the individual cargo unit of FIG. 1 in the collapsed configuration with the lid member of the second wall rotated to extend over the lid member of the first wall.

[0038] FIG. 6 is a perspective view of another cargo unit in an erect configuration.

[0039] FIG. 7 is a bottom plan view of the cargo unit of FIG. 6.

[0040] FIG. 8 is a side elevation partial cross-sectional view along the lines 8-8 of FIG. 7.

[0041] FIG. 9 is a rear elevation partial cross-sectional view along the lines 9-9 of FIG. 7.

[0042] FIG. 10 is a rear elevation view of the cargo unit of FIG. 6 in a collapsed configuration.

[0043] FIG. 11 is a rear elevation view of multiple collapsed cargo units in a stacked configuration.

DETAILED DESCRIPTION

[0044] With reference to the accompanying FIGS. 1 and 2, there is provided a collapsible, stackable cargo unit **20** having an erected configuration in which the cargo unit **20** may define an enclosed interior, and a collapsed configuration in which the cargo unit **20** may be stored and transported compactly. The cargo unit **20** comprises a generally rectangular base **22** having a front **24**, rear and two sides **26**, with forklift slots **28** on its front, rear and each side; front **30** and rear **32** walls extending upward from the base **22**; first and second side walls **34** and **36** extending upward from the base **22** between the front and rear walls; **30**, **32** and an integrated lid hingedly attached to one or more of the front, rear and side walls.

[0045] In some embodiments, all four walls are fully removable.

[0046] In some embodiments, the integrated lid comprises a first lid member **38** that includes a hinged edge **40** hingedly attached to the first side wall **34**, and a second lid member **42** that includes a hinged edge **44** hingedly attached to the second side wall **36**. In some embodiments, each of the first and second lid members **38**, **42** includes a free edge **46**, **48** opposite its hinged

edge, wherein the first lid member **38** is larger than the second lid member **42**, with a dimension between its hinged edge **40** and its free edge **46** being larger than a corresponding dimension of the second lid member **42**. In some embodiments, each lid member has a first handle **50** extending outward over the front wall **30**, and a second handle **52** extending outward over the rear wall **32**, to facilitate lifting of the lid to open the cargo unit **20**. Each handle **50** may include a downwardly-projecting end that wraps downward over the top of its associated front or rear wall to add strength and rigidity to the structure, and/or to help maintain the lid members in closed position.

[0047] In some embodiments, each of the lid members is permanently attached to a side wall, and is capable of pivoting through a 270° arc between a generally horizontal closed position in which the free ends of the lid members are adjacent one another, and a generally vertical open position in which the lid member hangs against an outside surface of its associated side wall, when the cargo unit is erected.

[0048] In some embodiments, the front wall **30** comprises releasable locking mechanisms **54** that are capable of locking the cargo unit **20** in erected configuration. In some embodiments, the front wall **30** comprises a lower front wall **56** and an upper front wall **58** pivotably connected thereto, wherein the lower front wall **56** includes locking mechanisms, the upper front wall **58** includes locking mechanisms, and when the cargo unit **20** is in its erected configuration, the upper front wall **58** may be either (1) locked in an upright configuration associated with a fully closed condition of the cargo unit, or (2) unlocked and pivoted outward and down through an arc of about 180° to hang against an outer surface of the lower front wall **56** while the lower front wall **56** remains locked in place, to partially open the cargo unit **20** and facilitate access to the interior while providing containment for a lower portion of the cargo. In some embodiments, the rear wall **32** is identical or substantially identical to the front wall **30** as described above.

[0049] In some embodiments, each of the base and walls comprises a metal frame, and one or more generally flat, planar metal sheets or plates welded or otherwise attached to the frame. Each frame may include a generally rectangular perimeter, and one or more stiffeners within the perimeter. The metal may be, e.g., galvanized steel, and the cargo unit may have a tare weight of, e.g., 70 to 100 kg., 80 to 90 kg., or about 84 kg. In other embodiments, one or more of the base and walls comprises a unitary, one-piece stamped primary member incorporating integral structural contours to provide strength and rigidity, instead of comprising a frame combined with sheets or plates, and the cargo unit may have a tare weight significantly less than 80 kg.

[0050] FIGS. **3** and **4** illustrate a double cargo unit **60** comprising a pair of similar individual cargo units **62** joined to one another back-to-back, wherein each of the individual cargo units is similar to one of those described above, except that it does not include a rear wall. Thus, the individual cargo units in these embodiments are joined at their respective open sides. The individual cargo units may be joined by bridge plates, splice bars, or other structures joining one or more of their base members, side walls and lids together, and/or by other means such as fastening adjacent portions of the bases and/or side walls directly to one another using fasteners **64** such as bolts. In some embodiments, the bases together define a generally flat floor, without any significant upward projection or rim protruding upward at the joint between the bases of the individual cargo units. In some embodiments, increased stiffness and strength are provided by a long, narrow removable support **66** such as a steel plate, steel angle, or a tubular member of e.g., rectangular cross section, extends across the joint at lid height, joining all four of the upper side wall corners adjacent the joint, and providing a support for edges of the lid members adjacent the joint. This structural member may have a slot or recess **68** to accommodate downward projections of the lid handles.

[0051] In some embodiments, the integrated lid of each individual cargo unit comprises a first lid member that includes a hinged edge hingedly attached to the first side wall, and a second lid member that includes a hinged edge hingedly attached to a second side wall, with edges of the lid members overlying the front walls, side walls and support **66**.

[0052] With reference to FIGS. **5A-5F**, a method of unloading and collapsing a collapsible,

stackable cargo unit as described above and changing its configuration from that shown in FIG. 1 to that shown in FIG. 2 may comprise: [0053] a. opening first and second lid members **38**, **42** by pivoting each lid member through an arc of about 270° about a hinge that connects it to a respective one of the first and second side walls **34**, **36**, as shown in FIG. 5A; [0054] b. removing cargo from the cargo unit; [0055] c. releasing locking mechanisms **54** on the front wall **30**, and pivoting the front wall **30** downward and inward about a pivotable connection to the base through an arc of about 90° so that it rests on the base **22**, as shown in FIG. 5B; [0056] d. releasing locking mechanisms on the rear wall **32**, and pivoting the rear wall **32** downward and inward about a pivotable connection to the base through an arc of about 90° so that it rests on the front wall **30**, as shown in FIG. 5C, with the side walls **34**, **36** remaining upright, supported by base corner posts; [0057] e. pivoting a first side wall **34** downward and inward about a pivotable connection to the base through an arc of about 90° so that it rests on the rear wall **32**, as shown in FIG. 5D; [0058] f. pivoting a second side wall **36** downward and inward about a pivotable connection to the base through an arc of about 90° so that it rests on the first lid member **38**, as shown in FIG. 5E; and [0059] g. pivoting the second lid member **42** through an arc of about 180° about the hinge that connects it to the second side wall **36** so that the second lid member **42** member rests on the first lid member **38** rather than the second side wall **36**, as shown in FIG. 5F, thereby reducing the height of the collapsed cargo unit.

[0060] In some embodiments, methods of erecting and loading a collapsed cargo unit may comprise reversing the above steps. In some embodiments, methods of collapsing or erecting a double cargo unit as described above may include methods identical to those described above, except that the steps involving rear walls will not be applicable, and the methods involving the other walls will need to be carried out on each of the individual cargo units.

[0061] In some embodiments involving loading or unloading of the individual or double cargo units described above, any number of walls may be removed from the base prior to loading or unloading, and later replaced. This may facilitate access to the interior from one or more sides, which may be useful in some contexts involving, e.g. manual loading or unloading, and/or use of forklifts to load or unload.

[0062] A cargo unit **20** may include an identification component **70** comprising an RFID tag, bar code label, and/or other identifier in a slot in the base **22**. This positioning offers protection from damage due to contact between cargo units, and also offers some protection from contact with forklift tines or other potential sources of damage.

[0063] In some embodiments, the cargo unit **20** has dimensions of 1150×995×835 mm, which can help to provide a “one dimension fits all” feature.

[0064] As discussed in greater detail below with respect to cargo unit **100**, the length of the first side wall **34** (i.e., the vertical length when in the erect configuration) may be different than the length of the second side wall **36**. In this way, the first lid member **38** may lay flat (e.g., generally coplanar) with the first side wall **34** when in the collapsed configuration to reduce the height of the stacked walls.

[0065] Referring now to FIG. 6, another cargo unit **100** is shown. The cargo unit **100** may be the same as the cargo unit **20** of FIGS. 1-5F. For example, the cargo unit **100** comprises a generally rectangular base **122** having a front **124**, rear, and two sides **126**, with forklift slots **128** on its front, rear, and each side; front **130** and rear **132** walls extending upward from the base **122**; first and second side walls **134** and **136** extending upward from the base **122** between the front and rear walls **130**, **132**; and an integrated lid hingedly attached to one or more of the front, rear and side walls. As discussed in greater detail with respect to FIG. 8, the first side wall **134** may have a length that is shorter than the length of the second side wall **136**.

[0066] In some embodiments, all four walls are fully removable.

[0067] In some embodiments, the integrated lid comprises a first lid member **138** that includes a hinged edge **140** hingedly attached to the first side wall **134**, and a second lid member **142** that

includes a hinged edge **144** hingedly attached to the second side wall **126**. In some embodiments, each of the first and second lid members **138**, **142** includes a free edge **146**, **148** opposite its hinged edge, wherein the first lid member **138** is larger than the second lid member **142**, with a dimension between its hinged edge **140** and its free edge **146** being larger than a corresponding dimension of the second lid member **142**. In some embodiments, each lid member has a first handle **150** extending outward over the front wall **130**, and a second handle **152** extending outward over the rear wall **132**, to facilitate lifting of the lid to open the cargo unit **100**. Each handle may include a downwardly-projecting end that wraps downward over the top of its associated front or rear wall to add strength and rigidity to the structure, and/or to help maintain the lid members in closed position.

[0068] In some embodiments, each of the lid members **138**, **142** is permanently attached to a side wall, and is capable of pivoting through a 270° arc between a generally horizontal closed position in which the free ends of the lid members are adjacent one another, and a generally vertical open position in which the lid member hangs against an outside surface of its associated side wall, when the cargo unit **100** is erected.

[0069] In some embodiments, the front wall **130** comprises releasable locking mechanisms **154** that are capable of locking the cargo unit in erected configuration. In some embodiments, the front wall **130** comprises a lower front wall **156** and an upper front wall **158** pivotably connected thereto, wherein the lower front wall **156** includes locking mechanisms, the upper front wall **158** includes locking mechanisms, and when the cargo unit **100** is in its erected configuration, the upper front wall **158** may be either (1) locked in an upright configuration associated with a fully closed condition of the cargo unit **100**, or (2) unlocked and pivoted outward and down through an arc of about 180° to hang against an outer surface of the lower front wall **156** while the lower front wall **156** remains locked in place, to partially open the cargo unit **100** and facilitate access to the interior while providing containment for a lower portion of the cargo. In some embodiments, the rear wall **132** is identical or substantially identical to the front wall **130** except that it may have a different height to accommodate vertically offset pivot axes so that when erect, the top edges of the front and rear walls will be coplanar while their bottom edges are at slightly different elevations.

[0070] In some embodiments, each of the base and walls comprises a metal frame, and one or more generally flat, planar metal sheets or plates welded or otherwise attached to the frame. Each frame may include a generally rectangular perimeter, and one or more stiffeners within the perimeter. The metal may be, e.g., galvanized steel, and the cargo unit may have a tare weight of, e.g., 70 to 100 kg., 80 to 90 kg., or about 84 kg. In other embodiments, one or more of the base and walls comprises a unitary, one-piece stamped primary member incorporating integral structural contours to provide strength and rigidity, instead of comprising a frame combined with sheets or plates, and the cargo unit may have a tare weight significantly less than 80 kg.

[0071] Referring to FIG. 7, the base **122** of the cargo unit **100** includes a lower front frame **160**, a lower rear frame **162** opposite the lower front frame **160**, a first lower side frame **164**, and a second lower side frame **166** opposite the first lower side frame **164**. The lower frames extend between corner posts **170** of the base **122**. The lower front and rear frames **160**, **162** have a common length that is longer than a common length of the first and second lower side frames **164**, **166**. For example, the lower front and rear frames **160**, **162** may each have a length that is approximately 15% longer than the lengths of the first and second lower side frames **164**, **166**. In this way, the lower frames cooperate to form a rectangular base **122**. The base **122** also includes support structures **172** that extend between the lower front and rear frames **160**, **162**, and transverse support structures **174** that extend between the first and second lower side frames **164**, **166**. The transverse support structures **174** may extend orthogonally relative to the support structures **172**. The support structures **172** and the transverse support structures **174** cooperate to support a floor **176** of the base **122**.

[0072] Referring to FIG. 8, the base **122** includes a first upright side portion **180** that extends

vertically between the first lower side frame **164** and a hinged edge **184** of the first side wall **134**, and a second upright side portion **182** that extends vertically between the second lower side frame **166** and a hinged edge **186** of the second side wall **136**. The upright side portions **180**, **182** may be fixed upright walls. For example, the upright side portions **180**, **182** may be integrally formed with the base **122**.

[0073] The upright side portions **180**, **182** of the base **122** may have different vertical lengths. For example, as shown in FIG. **8**, the first upright side portion **180** extends to a first vertical height, indicated at **190**, and the second upright side portion **182** extends to a second vertical height, indicated at **192**, that is vertically above (i.e., higher from a ground surface) the first vertical height **190**.

[0074] The first and second side walls **134**, **136** may be hingedly secured to the base adjacent the first and second upright side portions **180**, **182**. More particularly, the hinged edge **184** of the first side wall **134** is hingedly attached to a corner post **170** of the base **122** such that the first side wall **134** is pivotable about a first pivot axis **194**. The hinged edge **186** of the second side wall **136** is hingedly attached to a corner post **170** of the base **122** such that the second side wall **136** is pivotable about a second pivot axis **196** that is vertically offset from the first pivot axis **194**. As shown, the second pivot axis **196** is vertically above (i.e., higher from a ground surface) the first pivot axis **194**.

[0075] As discussed, the first and second side walls **134**, **136** may have different lengths. As used herein, a length of a side wall refers to a vertical dimension when the side wall is in an upright position, and a horizontal dimension when the side wall is in a collapsed position. As shown, the first side wall **134** may have a length, indicated at **200**, that is greater than a length of the second side wall **136**, indicated at **202**.

[0076] In this way, the vertical height **190** of the first upright side portion **180** and the length **200** of the first side wall **134** combine to form a combined length **204** of first side of the cargo unit **100**, and the vertical height **192** of the second upright side portion **182** and the length **202** of the second side wall **136** combine to form a combined length **206** of second side of the cargo unit **100**, such that the first combined length **204** is substantially the same as the second combined length **206**. As such, the two sides **126** of the cargo unit **100** extend to a substantially common height.

[0077] Referring to FIG. **9**, the front wall **130** has a length **210** substantially the same as the length **212** of the rear wall **132**.

[0078] The cargo unit **100** may be collapsed to the collapsed configuration shown in FIG. **10** in a manner similar to that discussed with respect to FIGS. **5A-5F**. For example, a user may pivot each of first and second lid members **138**, **142** through an arc of about 270° about a hinge that connects it to a respective one of the first and second side walls **134**, **136**. With the first and second lid members **138**, **142** resting vertically along first and second side walls **134**, **136** (similar to the configuration shown in FIG. **5A**), an operator may remove cargo from the cargo unit **100**.

[0079] A user may then release locking mechanisms **154** on the front wall **130**, and may pivot the front wall **130** downward and inward about a pivotable connection to the base **122** through an arc of about 90° so that it rests on the base **22** (similar to the configuration shown in FIG. **5B**).

[0080] Next, a user may then release locking mechanisms **154** on the rear wall **132**, and may pivot the rear wall **132** downward and inward about a pivotable connection to the base **122** through an arc of about 90° so that it rests on the front wall **130**, with the side walls **134**, **136** remaining upright, supported by base corner posts (similar to the configuration shown in FIG. **5C**).

[0081] A user may then pivot the first side wall **134** downward and inward about a pivotable connection to the base through an arc of about 90° so that it rests on the rear wall **132** (similar to the configuration shown in FIG. **5D**), and may then pivot the second side wall **136** downward and inward about a pivotable connection to the base **122** through an arc of about 90° so that it rests on the first lid member **138** (similar to the configuration shown in FIG. **5E**).

[0082] To reduce the height of the cargo unit **100** in the collapsed configuration, a user may pivot

the second lid member **142** through an arc of about 180° about the hinge that connects it to the second side wall **136** so that the second lid member **42** member rests on the first lid member **138** rather than the second side wall **136**.

[0083] As such, and as shown in FIG. **10**, in the collapsed configuration, the front wall **130** contacts and is supported by the floor **176** of the base **122**, and the rear wall **132** contacts and is supported by the front wall **130**. The first side wall **134** contacts and is supported by the rear wall **132**, with the first lid member **138** folded over and supported by the first side wall **134**. A distal end **220** of the second side wall **136** contacts and is supported by the first lid member **138**, with a proximal end **222** being spaced apart from the rear wall **132**. The second lid member **142** is rotated such that it contacts and is supported by the first lid member **138**. As shown, the combined length of the second side wall **136** and second lid member **142** corresponds to, or is less than, an internal dimension (e.g., an internal length) of the cargo unit **100**. In this way, the second side wall **136** and the second lid member **142** may be at least partially coplanar without interfering with opposite surfaces, such as posts **170**. In another approach, the combined length of the second side wall **136** and the second lid member **142** may exceed an internal dimension (e.g., an internal length) of the cargo unit **100**. In such an approach, the second lid member **142** may be dimensioned to pass between posts **170** so as not to interfere with the planar orientation of the second lid member **142**.

[0084] According to one aspect, the combined length of the first side wall **132** and the first lid member **138** may exceed an internal dimension (e.g., an internal length) of the cargo unit **100**. According to another aspect, the combined length of the first side wall **132** and the first lid member **138** may exceed an internal dimension (e.g., an internal length) of the cargo unit **100**.

[0085] In this way, folding of the front wall **130**, the rear wall **132**, the first side wall **134**, the first lid member **138**, and the second side wall **136** with the second lid member **142** being unfolded therefrom provides for a compact collapsed configuration of the cargo unit **100**. More particularly, the height of the stacked walls and lid members is reduced by the thickness of the second lid member **142**, which may be, for example, approximately 26 mm. Such a compact configuration allows for compact stacking of multiple cargo units **100**.

[0086] For example, referring to FIG. **11**, multiple cargo units **100**, **100'** may be stacked. In a stacked configuration, lower frame structures of an upper cargo unit **100'** are spaced (e.g., vertically spaced) from the stacked walls and lids of a lower cargo unit **100**. More particularly, the lower rear frame **162'** of the base **122'** of the upper cargo unit **100'** is spaced from the second side wall **136** and second lid member **142** of the lower cargo unit **100**. In the stacked configuration, an upper portion **230** of a corner post **170** of the lower cargo unit **100** may receive (or may be received within) a receptacle portion **232'** of a lower portion **234'** of an upper cargo unit **100'**.

[0087] From the foregoing, it should be appreciated that there is described and illustrated herein a complete metallic cargo unit with **4** doors fully removable, integrated lids that are foldable with associated walls, and foldable half doors with latches, that is returnable in one unit and individually and uniquely identified and traceable. In some embodiments, the cargo unit may be able to provide maximum capacity utilization within trucks, trailers, railcars, and ISO 40 ft. containers (e.g., one dimension fits all.)

[0088] It is contemplated that in various embodiments, any number of the features described herein may be included in combination with one another, to the extent practical. It is further contemplated that in various embodiments, any number of the features described in the above-mentioned U.S. Pat. No. 8,573,427 may be included in combination with one or more of the features described herein, to the extent practical.

[0089] The present disclosure describes embodiments and examples of cargo units and related methods of use and operation. Terms such as front, rear, side, vertical, horizontal, upper, lower, etc. are descriptive of the figures presented herein.

[0090] Those skilled in the art will recognize that a wide variety of modifications, alterations, and combinations can be made with respect to the embodiments described above without departing

from the scope as set forth in the claims, and that such modifications, alterations, and combinations are to be viewed as being within the ambit of the inventive concept. In addition, it should also be understood that features of one embodiment described herein may be combined with features of other embodiments described herein to provide yet other embodiments as desired.

Claims

1. A collapsible, stackable cargo unit having an erected configuration in which the cargo unit may define an enclosed interior, and a collapsed configuration in which the cargo unit may be stored and transported compactly, the cargo unit comprising: a base having a front, a rear and two sides; front and rear walls extending upward from the base; first and second side walls extending upward from the base between the front and rear walls, the first side wall having a length greater than the second side wall; an integrated lid hingedly attached to one or more of the front, rear, and side walls; and corner posts, wherein the front, rear, first side, and second side walls, and the integrated lid are positioned below the corner posts in the collapsed configuration to permit the cargo unit to stack upon another cargo unit.
2. The cargo unit of claim 1 wherein the base includes a first fixed upright wall adjacent to the first side wall and a second fixed upright wall adjacent to the second side wall, wherein the second fixed upright wall extends a greater height than the first fixed upright wall.
3. The cargo unit of claim 1 wherein the integrated lid comprises a first lid member that includes a hinged edge hingedly attached to the first side wall, and a second lid member that includes a hinged edge hingedly attached to the second side wall.
4. The cargo unit of claim 3 wherein in a collapsed configuration, the first lid member is rotated to rest on the first side wall the second lid member is generally coplanar with the second side wall.
5. The cargo unit of claim 4 wherein each of the first and second lid members includes a free edge opposite its hinged edge, and wherein the first lid member is larger than the second lid member, with a dimension between its hinged edge and its free edge being larger than a corresponding dimension of the second lid member.
6. The cargo unit of claim 5 wherein each lid member has a first handle extending outward over the front wall, and a second handle extending outward over the rear wall.
7. The cargo unit of claim 6 wherein each of the lid members is permanently attached to a side wall, and is capable of pivoting through a 270° arc between a generally horizontal closed position in which the free ends of the lid members are adjacent one another, and a generally vertical open position in which the lid member hangs against an outside surface of its associated side wall.
8. The cargo unit of claim 7 wherein the front wall comprises releasable locking mechanisms that are capable of locking the cargo unit in erected configuration.
9. The cargo unit of claim 8 wherein the front wall comprises a lower front wall and an upper front wall pivotably connected thereto, wherein the lower front wall includes locking mechanisms, the upper front wall includes locking mechanisms, and when the cargo unit is in its erected configuration, the upper front wall may be either (1) locked in an upright configuration associated with a fully closed condition of the cargo unit, or (2) unlocked and folded outward and down through an arc of about 180° to hang against an outer surface of the lower front wall while the lower front wall remains locked in place to partially open the cargo unit and facilitate access to the interior.
10. The cargo unit of claim 9 wherein each of the front, rear and side walls is fully removable from the base.
11. The cargo unit of claim 10 wherein each of the base and walls comprises a metal frame, and one or more sheets of metal welded or otherwise attached to the frame.
12. The cargo unit of claim 10 wherein one or more of the base and walls comprises a unitary, one-piece stamped primary member incorporating integral structural contours to provide strength and

rigidity.

13. The cargo unit of claim 1 wherein the cargo unit is made of galvanized steel, and has a tare weight of about 80 to 90 kg.

14. A double cargo unit having an erected configuration in which the cargo unit may define an enclosed interior, and a collapsed configuration in which the cargo unit may be stored and transported compactly; the double cargo unit comprising a pair of individual cargo units joined to one another; each of the individual cargo units comprising, when in erected configuration: a base having four sides, with forklift slots on at least one side; first, second and third upstanding walls extending upward from the base to define a three sided enclosure and an open side; an integrated lid hingedly attached to one or more of the walls; corner posts, wherein the first, second, and third upstanding walls, and the integrated lid are positioned below the corner posts in the collapsed configuration to permit the double cargo unit to stack upon another cargo unit; and the individual cargo units being joined together at their open sides.

15. The double cargo unit of claim 14 wherein the integrated lid of each individual cargo unit comprises a first lid member that includes a hinged edge hingedly attached to the first wall, and a second lid member that includes a hinged edge hingedly attached to a second wall.

16. The double cargo unit of claim 15 wherein the first upstanding wall of one of the individual cargo units comprises a lower front wall and an upper front wall pivotably connected thereto, wherein the lower front wall includes locking mechanisms, the upper front wall includes locking mechanisms, and when the double cargo unit is in its erected configuration, the upper front wall may be either (1) locked in an upright configuration associated with a fully closed condition of the cargo unit, or (2) unlocked and folded outward and down through an arc of about 180° to hang against an outer surface of the lower front wall while the lower front wall remains locked in place to partially open the cargo unit and facilitate access to the interior.

17. The double cargo unit of claim 16 wherein each of the upstanding walls is fully removable from the base.

18. The double cargo unit of claim 17 wherein each of the base and upstanding walls comprises a metal frame, and one or more sheets of metal welded or otherwise attached to the frame.

19. The double cargo unit of claim 18 wherein each of the base and walls comprises a unitary, one-piece stamped primary member incorporating integral structural contours to provide strength and rigidity.
